

Hamza Cherkaoui

☎ +33 7 62 92 68 88 ✉ hamza.cherkao@gmail.com
🌐 hcherkaoui.github.io 📄 [Hamza Cherkaoui](#) (h 8 - i10 8) 📄 [hcherkaoui](#) in [Hamza Cherkaoui](#)

Interests

Bandits; Diffusion models; Markov chains; LLMs; Curriculum learning

Research experience

2025–now Post doctoral position, Télécom SudParis, Palaiseau, France

Post doctorate position

- *Collaborators*: Prof. Hélène Halconruy.
- I focus my research on the collaboration in dynamic problem.
- Fields of interest: *bandits, Bayesian optimization, Markov chains, Diffusion models, LLMs.*
- 1 pre-print, first author.

2022–2025 Post doctoral position, Huawei Technologies, Paris, France

Post doctorate position

- *Collaborators*: Dr. Igor Colin and Dr. Merwan Barlier.
- I focus my research on the multi-agents linear bandits problems.
- Fields of interest: *multi-/single agent(s) linear bandits, Bayesian optimization, optimization, LLMs.*
- 4 pre-print, first author, under review.

2021 Post doctoral position, BioMaps research team, CEA-Saclay, Orsay, France

Post doctorate position

- I continue my research on the estimation of the Haemodynamic Response Function (HRF) from fMRI data in a pharmacological context.
- Published 1, first author, International Conference Proceedings.

2017–2021 PhD program, PARIETAL research team, CEA-Saclay / INRIA-Saclay, France

PhD program

- *Research subject*: ‘Efficient whole brain estimation of the haemodynamic response function for TV-regularized semi-blind deconvolution of neural activity in fMRI’
- *Supervisors*: Dr. Philippe Ciuciu, Dr. Thomas Moreau and Dr. Claire Leroy
- Published 5, first author, International Conference Proceedings and 1, first author, Journal Articles.

Education

2016–2017 Master program, Ecole Polytechnique, Palaiseau, France

- *Master of Research*: Data Sciences
- Cross disciplinary courses: Machine Learning Theory, Kernel methods, Convex Optimization.

2015–2016 Master program, University Lille 1, Villeneuve d’Ascq, France

- *Master of Research*: Applied mathematics
- Focused on Probability and Statistic.

2012–2016 Ecole Centrale de Lille, Villeneuve d’Ascq, France

- *Master of Engineering*: diplôme d’ingénieur de l’Ecole Centrale de Lille.
- Cross disciplinary courses centered around mathematics and computer science.

Other Experiences

2017 Researcher assistant (internship), PARIETAL research team, Neurospin laboratory CEA-Saclay, France

April–September

- I optimized the reconstruction process of MRI images to enhance image quality with Compressed Sensing tools.

2016 Research engineer (internship), MODAL research team, INRIA Lille, France

June–August

- *Software development project*: Development of a new R package providing tools to cluster categorical functions with on an efficient C++ back-end implementation.

2015 Software developper (internship), DataRobot, Boston, USA

April–July

- *Software development project*: Implementation of the decision tree induction algorithm (from Scikit-Learn) used within the start-up to support multi-threading.

2014 Software developper (internship), Logilab, Paris, France

May–December

- *Software development project*: Implementation of a Python library based on CGAL library (Python/C++ binding) & development of a web-client toolkit to use a computation platform.

Languages & Computer Science skills

- *Languages*: French (mother-tongue) English (TOEIC 970/990, fluent)
- *Collaborative tools*: Git
- *Softwares*: Python (knowledge in Cython, C, C++)
- *Libraries*: Numpy, Pandas, Numba, pyTorch, pyTorch-Geometric, Scipy, Scikit-learn, ...

First author publications

- [1] Cherkaoui, H. and Barlier, M. and Colin, I. *Adaptive Sample Sharing for Multi Agent Linear Bandits*, International Conference on Machine Learning, 2025.
- [2] Cherkaoui, H. and Tiomoko, M. and Seddik, M. and Louart, C. and Schnoor, E. and Kegl, B. *High-Dimensional Analysis of Bootstrap Ensemble Classifiers*, Pre-print under review, 2025.
- [3] Cherkaoui, H. and Colin, I. and Shang, X. and Barlier, M. *Price of Safety in Linear Best Arm Identification*, Pre-print under review, 2023.
- [4] Cherkaoui, H. and Moreau, T. and Halimi, A. and Leroy, C. and Ciuciu, P. *Multivariate semi-blind deconvolution of fMRI time series*, NeuroImage, 2021.
- [5] Cherkaoui, H. *Efficient whole brain estimation of the haemodynamic response function for TV-regularized semi-blind deconvolution of neural activity in fMRI*, Ph.D. thesis, 2021.
- [6] Cherkaoui, H. and Moreau, T. and Ciuciu, P. and Fernandez, B. and Bottlaender, M. and Tournier, N. and Leroy, C. *Characterization of the haemodynamic response function after a buprenorphine challenge study in Human healthy volunteer*, 27th Annual Meeting and Educational Courses of OHBM, 2021.
- [7] Cherkaoui, H. and Sulam, J. and Moreau, T. *Learning to solve TV regularised problems with unrolled algorithms*, 34th Conference and Workshop on Neural Information Processing Systems (NeurIPS), 2020.
- [8] Cherkaoui, H. and Moreau, T. and Halimi, A. and Ciuciu, P. *Sparsity-based blind deconvolution of neural activation signal in fMRI*, 44th International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2019.
- [9] Cherkaoui, H. and Moreau, T. and Halimi, A. and Ciuciu, P. *fMRI BOLD signal decomposition using a multivariate low-rank model*, 27th European Signal Processing Conference (EUSIPCO), 2019.
- [10] Cherkaoui, H. and Gueddari, L. and Lazarus, C. and Grigis, A. and Poupon, F. and Vignaud, A. and Farrens, S. and Starck, J.-L. and Ciuciu, P. *Analysis vs synthesis-based regularization for combined compressed sensing and parallel MRI reconstruction at 7 tesla*, 26th European Signal Processing Conference (EUSIPCO), 2018.

Developped Python packages

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| 2025 | Metis , Auto Machine Learning in Python. <ul style="list-style-type: none">• Designed for automated model selection and ensembling to optimize predictive performance. |
| 2023 | Bandpy , Multi-/single agent(s) bandits in Python. <ul style="list-style-type: none">• Designed to streamline research benchmarking while ensuring compliance with the GYM API.• Include the linear, quadratic and multi-armed bandits with the common associated algorithms. |
| 2020 | Carpet , Fast-minimization of Total Variation regularized problems.  carpet <ul style="list-style-type: none">• Minimize 1d TV regularized problems with neural network. |
| 2019 | HemoLearn , Multivariate estimation of the HRF from fMRI data.  hemolearn <ul style="list-style-type: none">• Disentangle the neurovascular coupling from the neural activation signal in a multivariate fashion for fMRI data. |
| 2018 | pyBOLD , Univariate estimation of the HRF from fMRI data.  pybold <ul style="list-style-type: none">• Disentangle the neurovascular coupling from the neural activation signal in a univariate fashion for fMRI data. |
| 2018 | pysap , MRI image reconstruction.  pysap <ul style="list-style-type: none">• MRI image reconstruction based on Compressed Sensing (CS). |